



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Divya Konar
Roll No. :	20
Experiment No.:	1
Title of the Experiment:	Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)
Date of Performance:	22/07/26
Date of Submission:	29/07/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi P.N.**

Signature :

Date :



Aim: To perform exploratory statistical analysis of the Pima Indians Diabetes Dataset by examining its structure, variable types, data quality, statistical characteristics, and important patterns.

Objectives

After completing this experiment, the student will be able to:

1. Apply exploratory data analysis techniques to understand the structure and characteristics of a real-world healthcare dataset.
2. Interpret the statistical characteristics, data quality issues, and important patterns identified during the analysis.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations, 8 input attributes, and one binary outcome variable.



Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Examine the dataset dimensions, column names, data types, and sample records.
3. Classify the variables as numerical, categorical, or binary.
4. Generate a statistical summary of the dataset.
5. Check for missing or invalid values.
6. Identify potential outliers using suitable statistical techniques and visualizations.
7. Generate appropriate visualizations to understand the data.
8. Record and interpret important observations obtained from the exploratory analysis.

Description of the Experiment

Exploratory Data Analysis (EDA) is the initial process of understanding a dataset before applying statistical inference or machine learning techniques.

The analysis follows: Raw Dataset, Data Inspection, Variable Identification, Data Quality, Analysis, Statistical Summary, Visualization and Interpretation

The Pima Diabetes Dataset is explored to understand its structure, data quality, variable distributions, and important characteristics before developing predictive models.

Detailed Description of Statistical techniques:

Descriptive Statistics

Statistical measures such as the following are used to summarize numerical variables:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard Deviation

Missing Value Analysis

The dataset is examined for missing or invalid values.

For example, zero values in certain medical attributes such as **Glucose**, **BloodPressure**, **SkinThickness**, **Insulin**, and **BMI** may require investigation because a zero value may represent missing or invalid data rather than a genuine medical measurement.



Outlier Analysis

Potentially unusual observations are identified using statistical summaries and visualizations such as boxplots.

An outlier should be investigated before deciding whether it should be removed or retained.

Data Visualization

Suitable visualizations are generated to understand:

- Variable distributions
- Frequency of diabetes outcomes
- Relationships between variables
- Potential outliers

Examples include:

- Histograms
- Boxplots
- Bar charts
- Scatter plots

Conclusion

Exploratory statistical analysis was performed on the Pima Indians Diabetes Dataset to understand its structure, variable characteristics, data quality, and important patterns. The analysis provides the foundation for further statistical analysis and machine learning model development.

Output :



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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcl...) Code

[Running] python -u "e:\MYDATA\STATS\eda_diabetes.py"

	Pregnancies	Glucose	BloodPressure	...	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	...	0.627	50	1
1	1	85	66	...	0.351	31	0
2	8	183	64	...	0.672	32	1
3	1	89	66	...	0.167	21	0
4	0	137	40	...	2.288	33	1

[5 rows x 9 columns]

Dataset Shape:
(768, 9)

Column Names:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
 dtype='str')

Data Types:
Pregnancies int64
Glucose int64
BloodPressure int64
SkinThickness int64
Insulin int64
BMI float64
DiabetesPedigreeFunction float64
Age int64
Outcome int64
dtype: object

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcl...) C

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)
memory usage: 54.1 KB

Statistical Summary:

	Pregnancies	Glucose	...	Age	Outcome
count	768.000000	768.000000	...	768.000000	768.000000
mean	3.845052	120.894531	...	33.240885	0.348958
std	3.369578	31.972618	...	11.760232	0.476951
min	0.000000	0.000000	...	21.000000	0.000000
25%	1.000000	99.000000	...	24.000000	0.000000
50%	3.000000	117.000000	...	29.000000	0.000000
75%	6.000000	140.250000	...	41.000000	1.000000
max	17.000000	199.000000	...	81.000000	1.000000

[8 rows x 9 columns]

Missing Values:

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0



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[8 rows x 9 columns]

Missing Values:

Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

Invalid (Zero) Values:

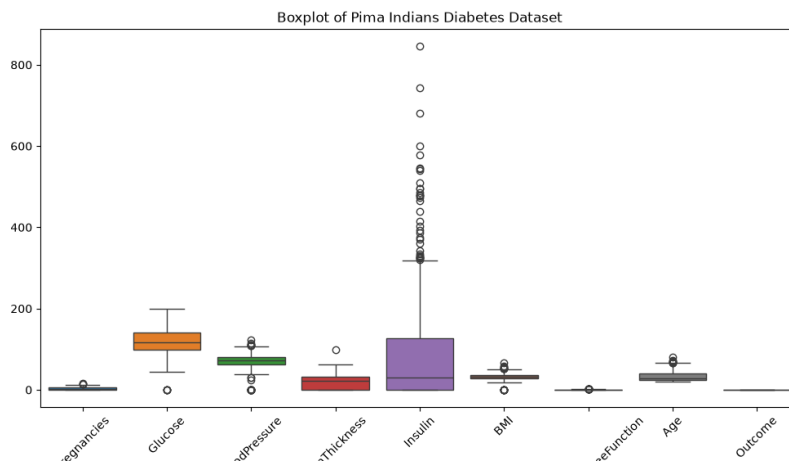
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

dtype: int64

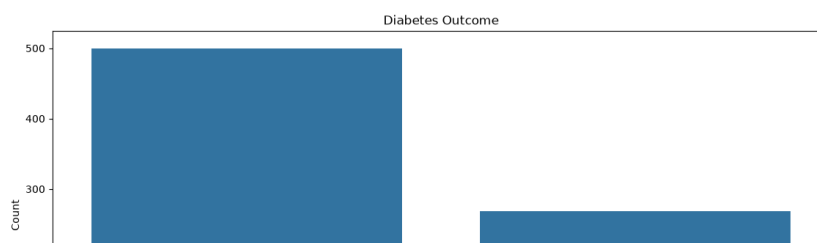
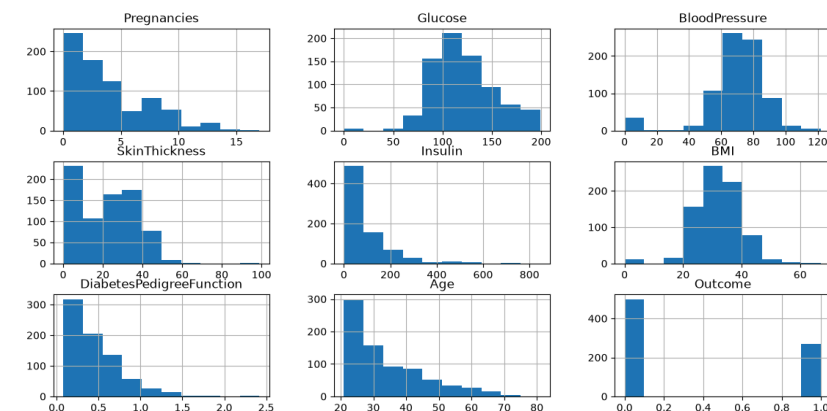
Number of Outliers:

Pregnancies	4
Glucose	5
BloodPressure	45
SkinThickness	1
Insulin	34
BMI	19
DiabetesPedigreeFunction	29
Age	9
Outcome	0

dtype: int64



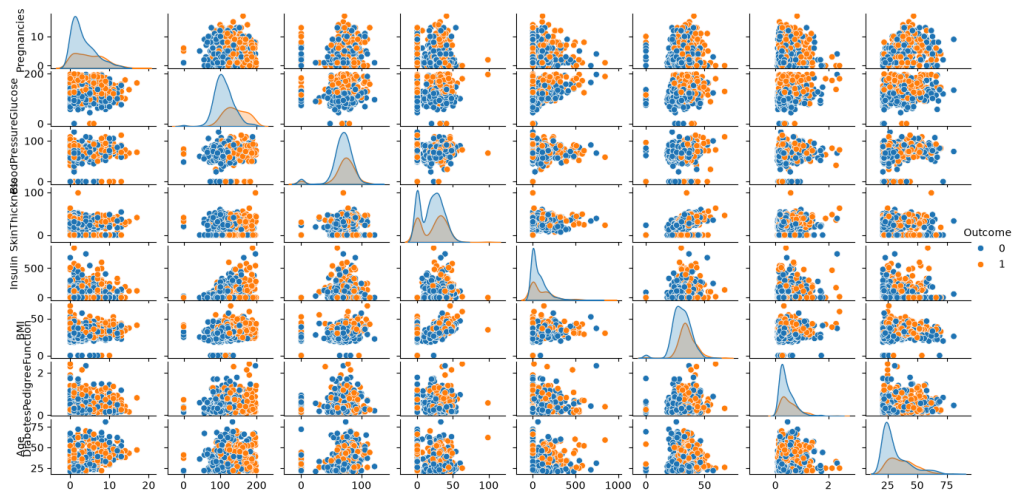
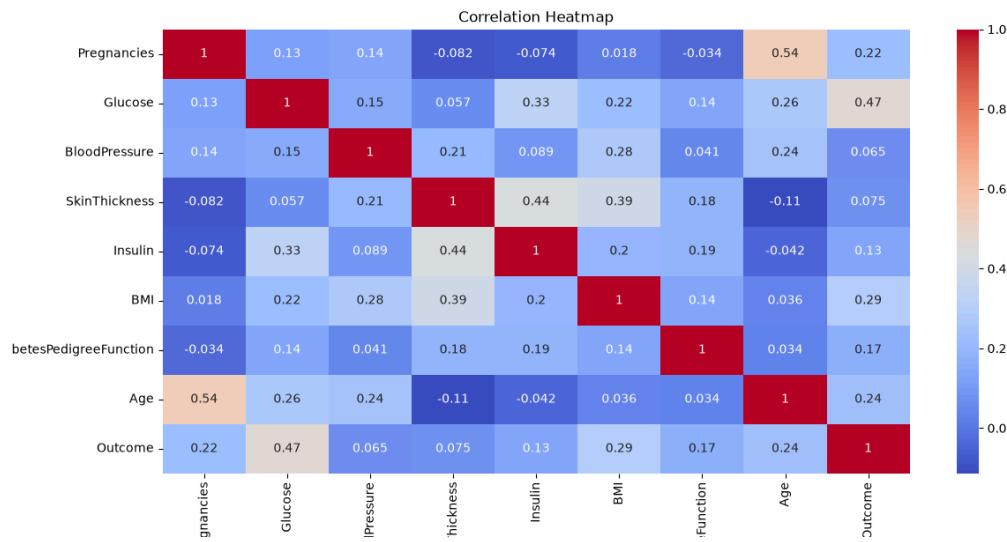
Histogram of Pima Indians Diabetes Dataset





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Questions to be Answered by Students

1. Which variables in the dataset contain zero or potentially invalid values that require investigation?

ANS : Glucose, BloodPressure, SkinThickness, Insulin, and BMI contain zero values, which are likely invalid and require investigation.

2. Which variable contains potential outliers based on the analysis?

ANS : Insulin contains the most significant outliers. BMI, SkinThickness, and DiabetesPedigreeFunction may also have some outliers.

3. What are the important patterns observed between the input variables and the Outcome variable?

ANS : People with diabetes (Outcome = 1) generally have higher Glucose, BMI, Age, and Insulin values than those without diabetes.

4. State **two important insights** obtained from the exploratory analysis.

ANS : 1. Glucose has the strongest relationship with diabetes.

2. Several variables contain invalid zero values and need preprocessing before model building.